

Nuvoton NuML_Studio Quick Start with Edge Impulse Project

Description

This document explains how to build a TinyML project with Edge Impulse and NuML_Studio, and deploy it on the NuMaker-M55M1 using image classification as an example.

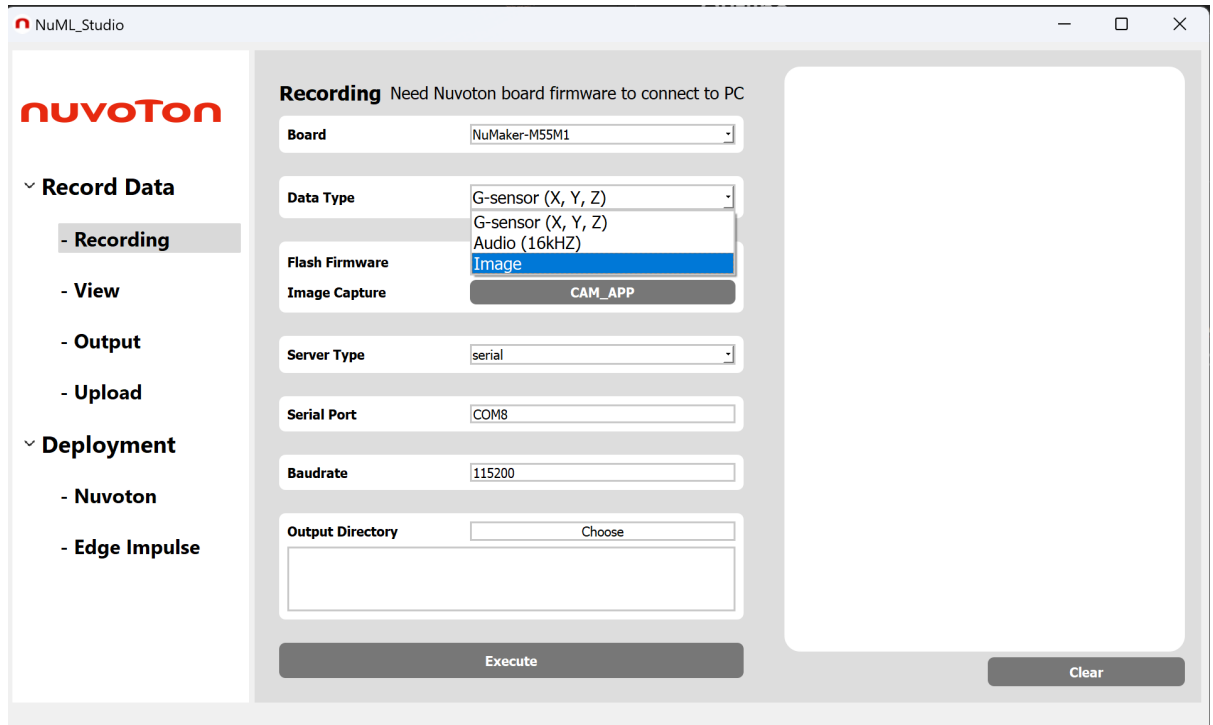
Outline

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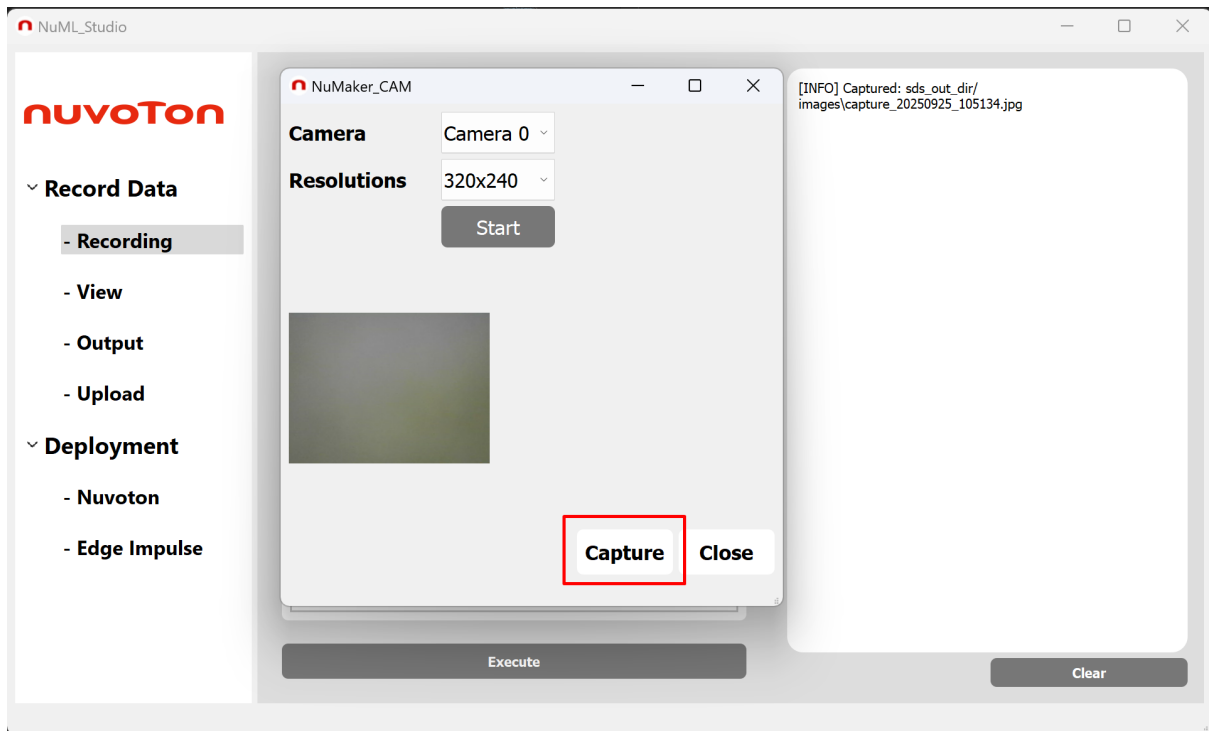
Getting Started

Collecting Data by NuML_Studio

1. Flash the data collection firmware
 - Select **Image**, click **Flash Firmware** and wait for completion.

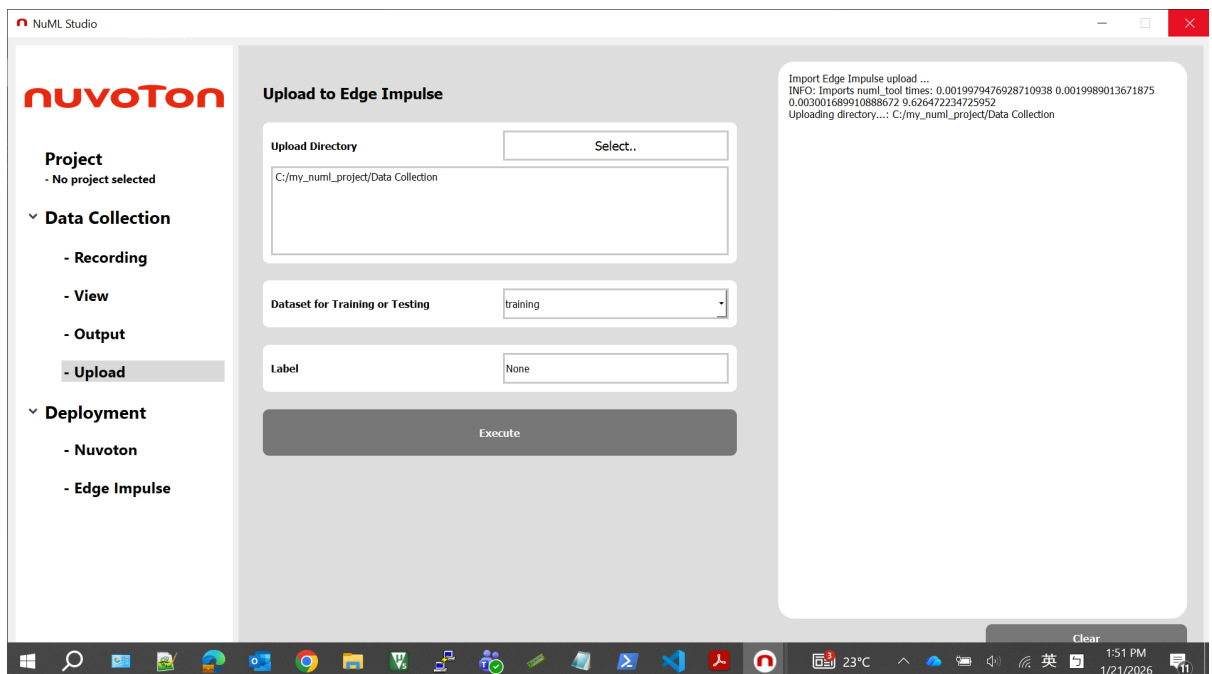


- Click **Image Capture** to take and save images.



2. Upload data to Edge Impulse

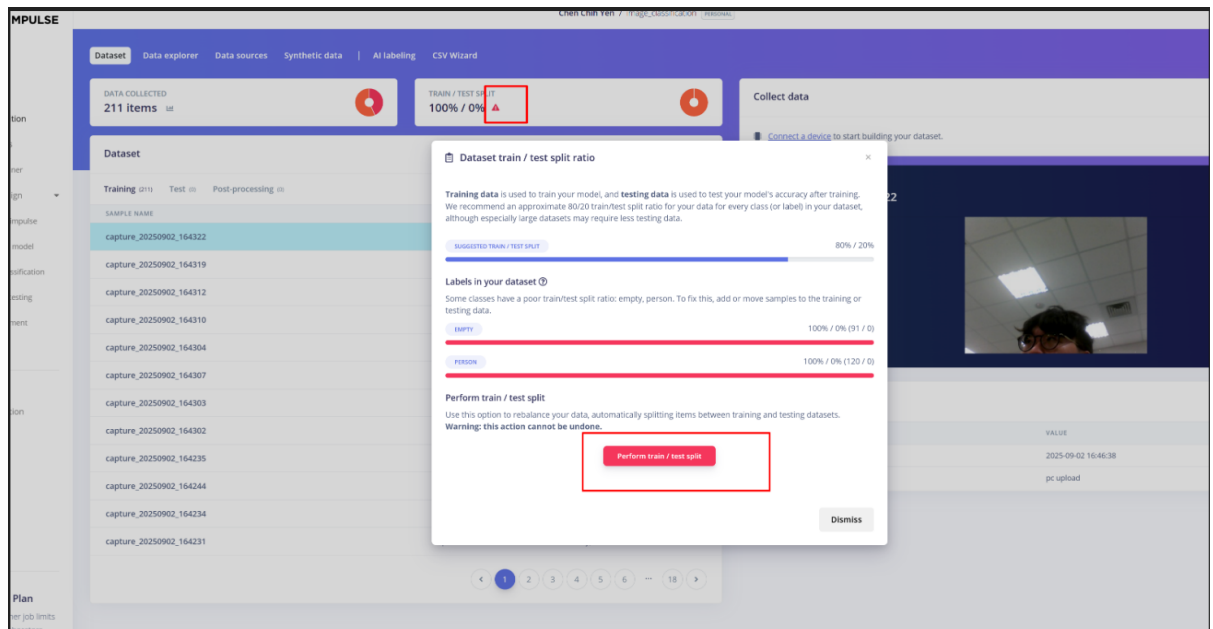
- Copy your project API-Key from Edge Impulse **Dashboard**, **Keys** and paste it into **NuML_Studio_vX.X.X\API_key.txt**.
- Use NuML_Studio to upload your collected data folder, or upload directly via Edge Impulse **Data acquisition** page.



Training by Edge Impulse

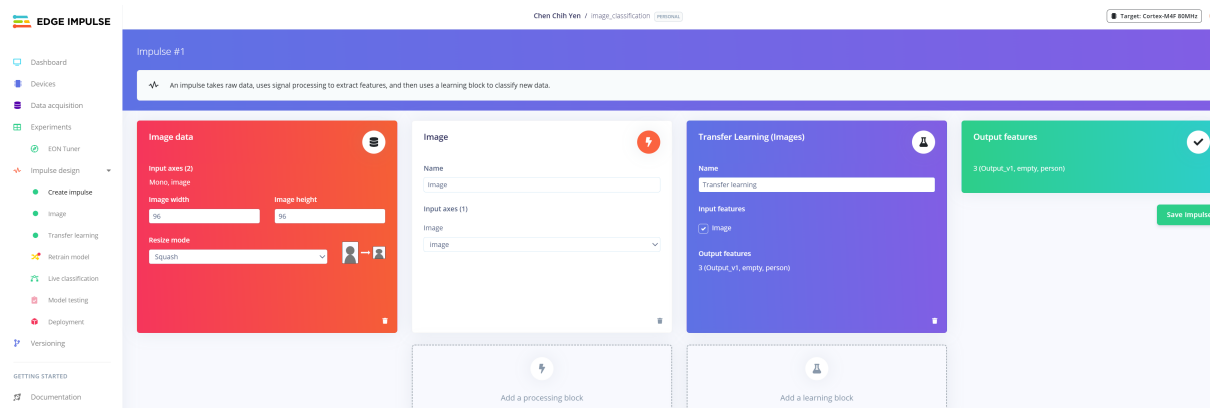
1. Data Acquisition

- Go to the **Data acquisition** tab in Edge Impulse.
- Review and update labels if needed.
- Split the collected data into training and testing datasets.



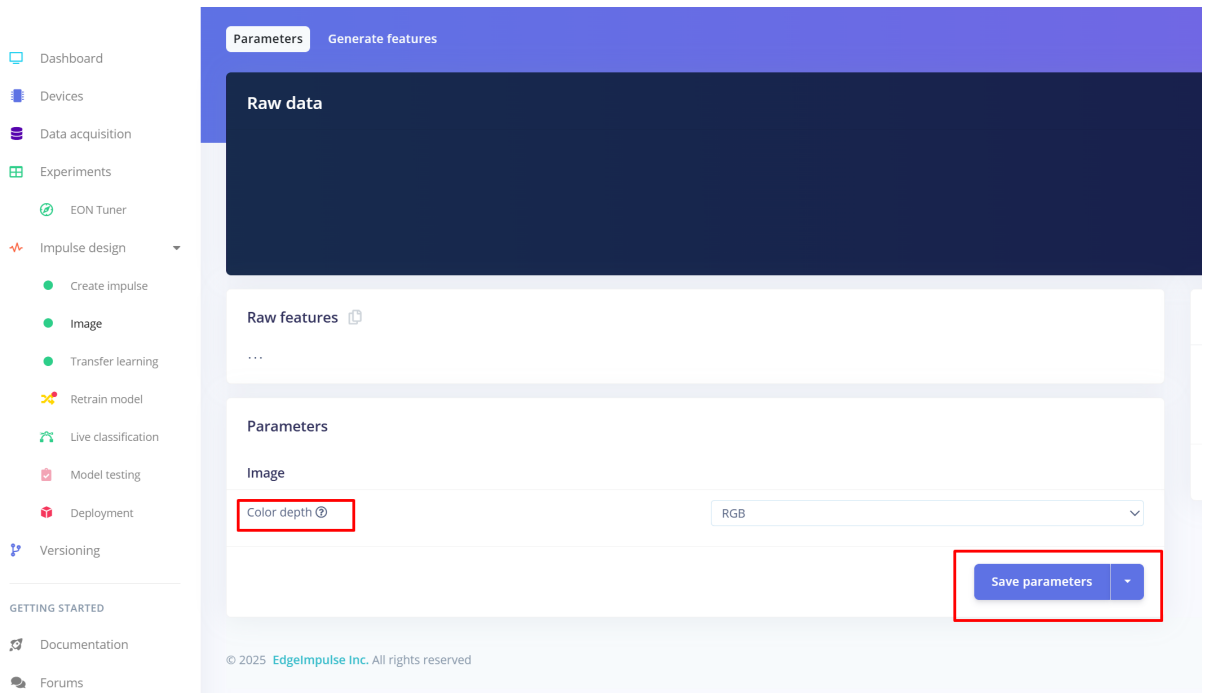
2. Create Impulse

- Go to the **Create Impulse** tab.
- Configure your ML pipeline (preprocessing, learning block, classifier).
- For Image input, set the **Resize mode** to **Squash**

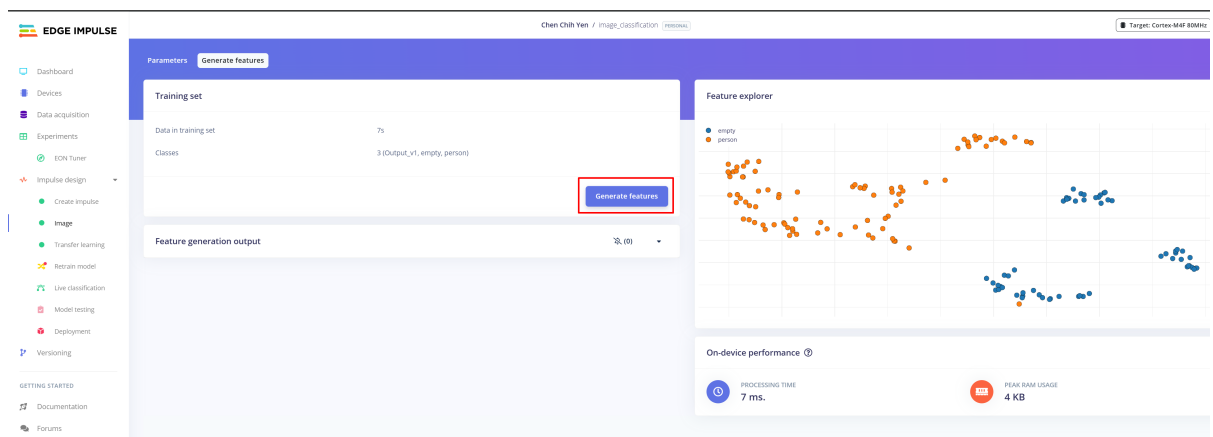


3. Set Parameters and generate Features

- Go to the **Image** tab.
- Set Parameters and click **Save parameters**.

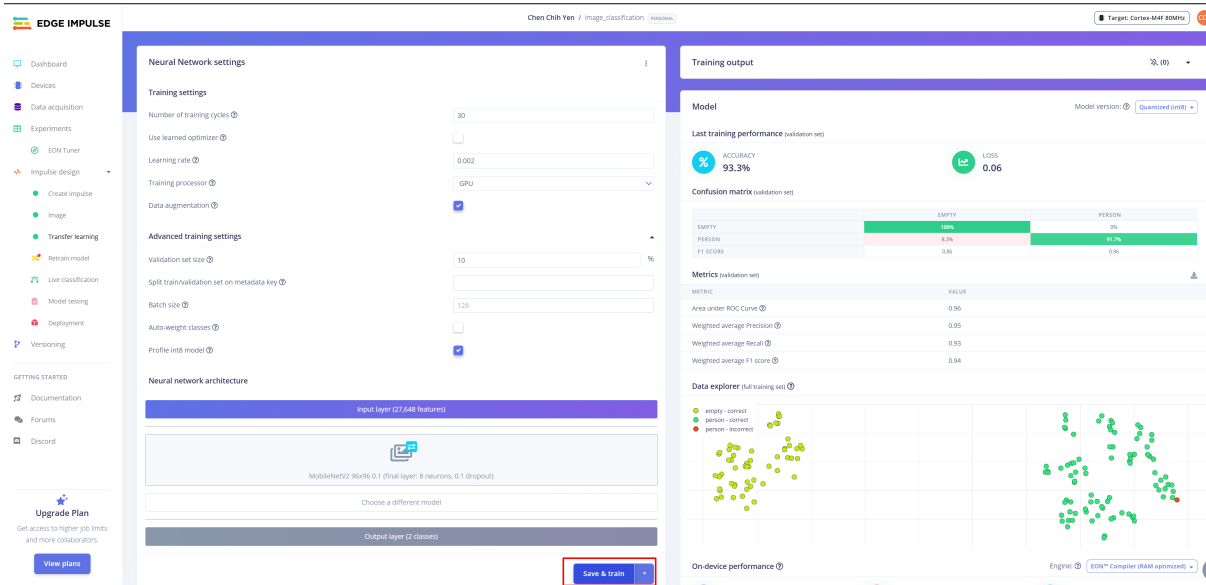


- Then click **Generate features** to extract feature vectors from your dataset.



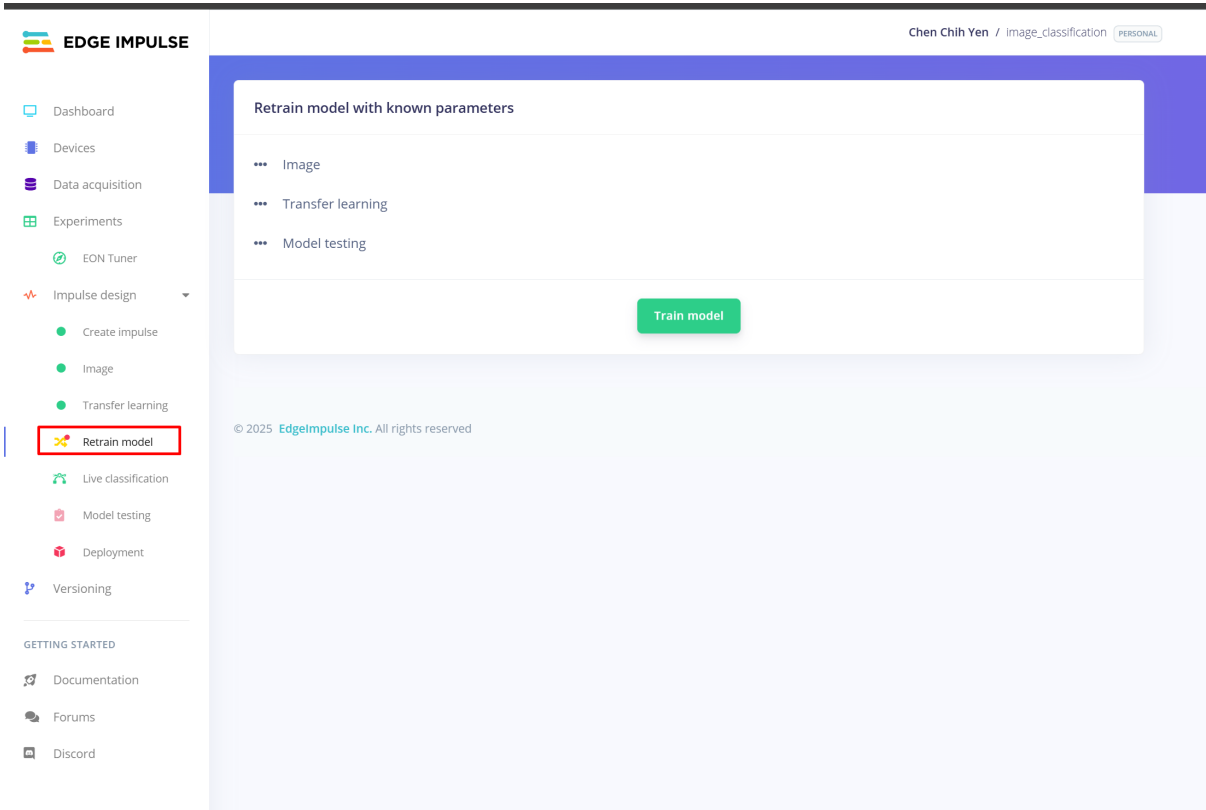
4. Start Training

- Go to the **Transfer Learning** or **Learning** tab.
- Select the model type.
- Adjust training parameters (epochs, learning rate, validation split, etc.).
- Click **Start training** to begin model training.



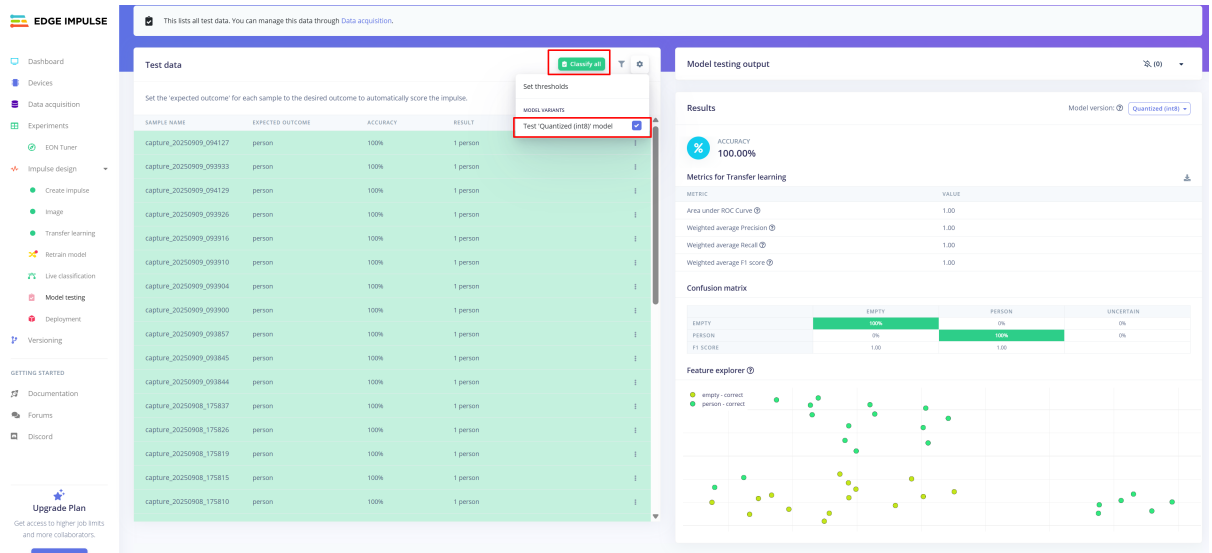
5. Retrain model

- Go to the Retrain model tab.



6. Test the model

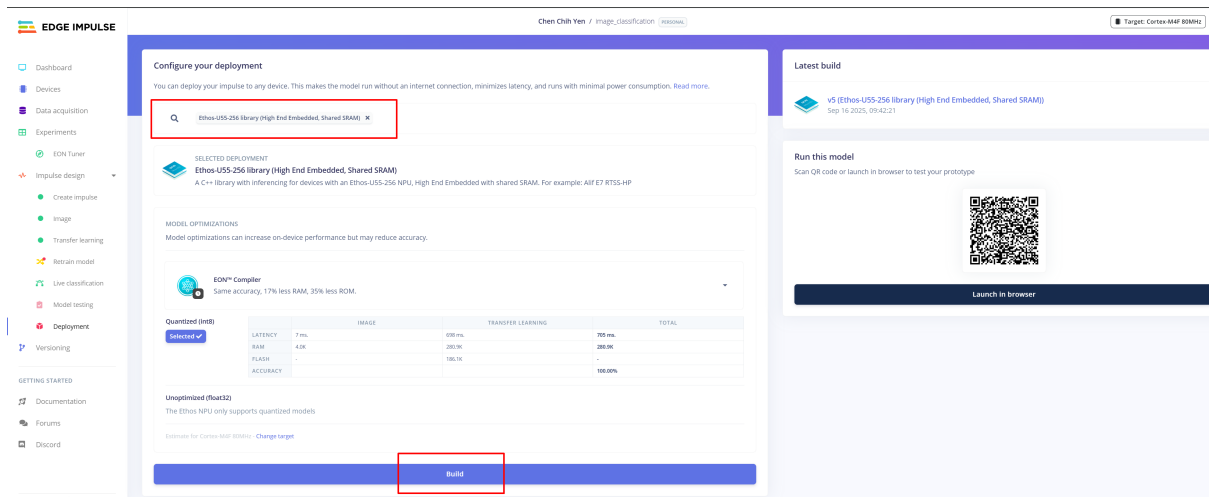
- Go to the Model testing tab.
- Please choose **Test Quantized model**. This will test the quantized model using all the test data.



Deployment

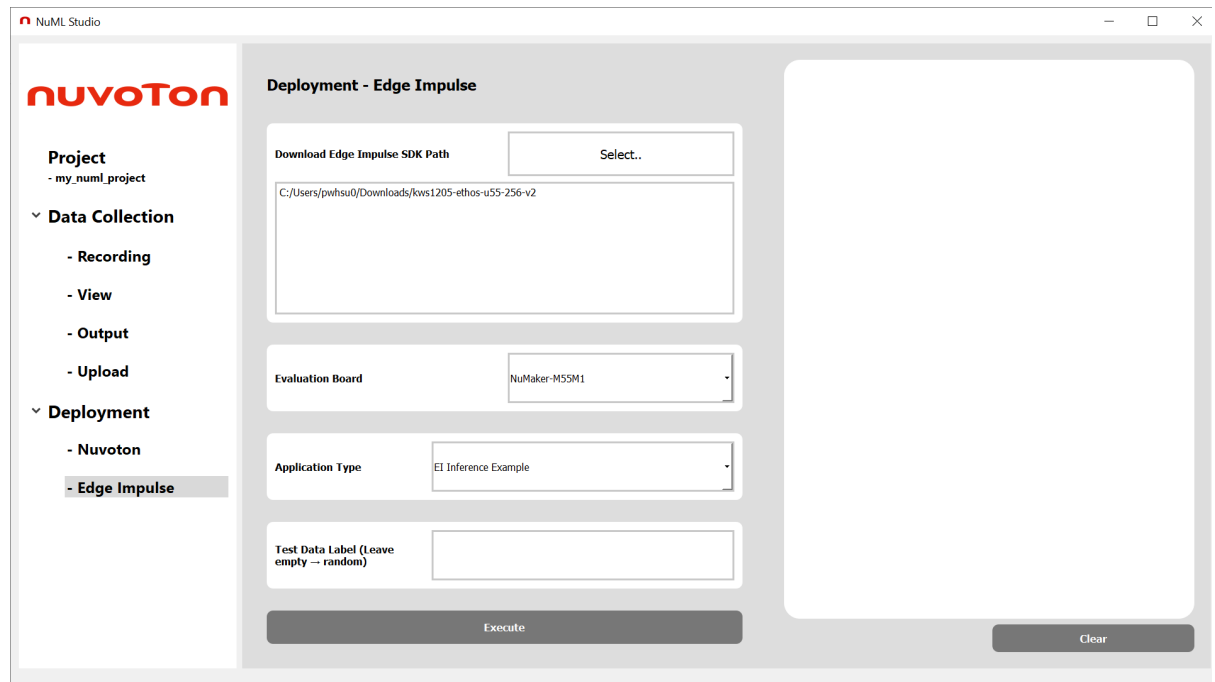
1. Build the deployment model and SDK

- Go to the **Deployment** tab on the Edge Impulse project page.
- Select **ethosu-U55 256 library** and build it.



2. Run inference firmware on NuMaker-M55M1

- After downloading and unzipping the SDK, use NuML-Studio to create the code project.
- In NuML-Studio, select - **Deployment**, - **Edge Impulse**.



- The final step is to open the Keil or VSCode CMSIS project generated by NuML-Studio, then build and flash the NuMaker-M55M1 ML firmware with your Edge Impulse model.